

Problem 1

```
import pandas as pd

from sklearn.linear_model import LinearRegression

from sklearn.model_selection import train_test_split

# Dataset

data = {

    'ctr': [3.2, 5.8, 2.1, 7.4, 4.5, 6.2, 1.8, 5.1, 3.9, 8.5,

            4.2, 2.8, 6.8, 3.5, 7.1, 2.4, 5.5, 4.8, 6.5, 3.1,

            7.8, 2.6, 5.3, 4.1, 6.9, 3.7, 8.1, 2.2, 5.9, 4.6],

    'total_views': [12000, 28000, 8500, 42000, 19000, 33000, 7000, 24000,

                    16000, 51000,

                    18000, 11000, 38000, 14500, 44000, 9500, 26000, 21000,

                    35000, 13000,

                    47000, 10000, 25000, 17500, 40000, 15000, 49000, 8000,

                    31000, 20000]

}

df = pd.DataFrame(data)

x=df[["ctr"]]

y=df["total_views"]

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
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model=LinearRegression()

model.fit(x_train,y_train)

print(f"the coefficient is {model.coef_}")

print(f"the intercept is {model.intercept_}")


y_pred=model.predict(x_test)

print("\nPredictions vs Actual:")

results = pd.DataFrame({'Actual': y_test.values, 'Predicted': y_pred.round()})

print(results)


new_ctr=pd.DataFrame([[8]],columns=["ctr"])

predictions=model.predict(new_ctr)

print("the predicted total_viwes",predictions)

from sklearn.metrics import r2_score


y_pred = model.predict(x)

score = r2_score(y, y_pred)

print(f"the accuracy of model is {score}")

import matplotlib.pyplot as plt

plt.scatter(x,y,color="blue",label="prev data")

plt.plot(x,y_pred,color="red",label="new data")
```

```

plt.xlabel("ctr", fontsize=16)

plt.ylabel("total_viwes", fontsize=16)

plt.title("ctr vs total_views")

plt.grid(True)

plt.tight_layout()

plt.legend()

plt.show()

```

problem2

```

from sklearn.metrics import r2_score

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

# Dataset

data = {

    'distance_km': [2.5, 6.0, 1.2, 8.5, 3.8, 5.2, 1.8, 7.0, 4.5, 9.2,

                    2.0, 6.5, 3.2, 7.8, 4.0, 5.8, 1.5, 8.0, 3.5, 6.8,

                    2.2, 5.5, 4.2, 9.0, 2.8, 7.2, 3.0, 6.2, 4.8, 8.2],

    'prep_time_min': [10, 20, 8, 25, 12, 18, 7, 22, 15, 28,

                      9, 19, 11, 24, 14, 17, 6, 26, 13, 21,

                      10, 16, 14, 27, 11, 23, 12, 18, 15, 25],

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        'delivery_time_min': [18, 38, 12, 52, 24, 34, 14, 45, 29, 58,
                               15, 40, 21, 50, 27, 35, 11, 54, 23, 43,
                               17, 32, 26, 56, 19, 47, 20, 37, 30, 53]
    }

df = pd.DataFrame(data)

print("the data")

print(df.head())

print("the stats ")

print(df.describe())

x=df[["distance_km", "prep_time_min"]]

y=df["delivery_time_min"]

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=
42)

model=LinearRegression()

model.fit(x_train,y_train)

y_pred=model.predict(x_test)

new_value=pd.DataFrame([[7,15]],columns=["distance_km", "prep_time_min"])

print(model.predict(new_value))

score=r2_score(y_test,y_pred)

print(score)

fig,axes=plt.subplots(1,2)

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scatter_1=axes[0].scatter(df["distance_km"],df["delivery_time_min"])

axes[0].set_xlabel("dis")

axes[0].set_title("dis vs time")

axes[0].set_ylabel("time")


scatter_2=axes[1].scatter(df["prep_time_min"],df["delivery_time_min"])

axes[1].set_xlabel("prep,time")

axes[1].set_ylabel("delivery,time")

axes[1].set_title("time and time")


plt.show()

```

problem 3

```

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import r2_score


# Dataset

data = {

    'ram_gb': [4, 8, 4, 16, 8, 8, 4, 16, 8, 16,

               4, 8, 4, 16, 8, 8, 4, 16, 8, 16,

```

```

        4, 8, 4, 16, 8, 8, 4, 16, 8, 16],

    'storage_gb': [256, 512, 128, 512, 256, 512, 256, 1024, 256, 512,
                    128, 512, 256, 1024, 256, 512, 128, 512, 256, 1024,
                    256, 512, 128, 512, 256, 512, 256, 1024, 256, 512],

    'processor_ghz': [2.1, 2.8, 1.8, 3.2, 2.4, 3.0, 2.0, 3.5, 2.6, 3.0,
                      1.6, 2.8, 2.2, 3.4, 2.5, 2.9, 1.9, 3.1, 2.3, 3.6,
                      2.0, 2.7, 1.7, 3.3, 2.4, 3.0, 2.1, 3.5, 2.6, 3.2],

    'price_inr': [28000, 45000, 22000, 72000, 38000, 52000, 26000, 95000,
                  42000, 68000,
                  20000, 48000, 29000, 88000, 40000, 50000, 23000, 70000,
                  36000, 98000,
                  25000, 46000, 21000, 75000, 39000, 53000, 27000, 92000,
                  43000, 73000]
}

#16GB RAM, 512GB storage, 3.2 GHz processor, what is a fair price?

df = pd.DataFrame(data)

print(df.head())

print(df.describe())

x=df[["ram_gb", "storage_gb", "processor_ghz"]]

y=df["price_inr"]

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=
42)

```

```
model=LinearRegression()

model.fit(x_train,y_train)

y_pred=model.predict(x_test)

new_values=pd.DataFrame([[16,512,3.2]],columns=
["ram_gb", "storage_gb", "processor_ghz"])

print(model.predict(new_values))

print(r2_score(y_test,y_pred))

fig,axes=plt.subplots(1,3,figsize=(15,5))

scatter1=axes[0].scatter(df["ram_gb"],df["price_inr"])

axes[0].set_xlabel("ram_gb")

axes[0].set_title("gb vs price")

axes[0].set_ylabel("price_inr")


scatter2=axes[1].scatter(df["storage_gb"],df["price_inr"])

axes[1].set_xlabel("storage_gb")

axes[1].set_title("storage vs price")

axes[1].set_ylabel("price_inr")


scatter3=axes[2].scatter(df["processor_ghz"],df["price_inr"])

axes[2].set_xlabel("processer_ghz")

axes[2].set_title("processer vs price")

axes[2].set_ylabel("price_inr")

plt.show()
```

